

Standardizing Safety Analyses for Clinical Trials Across the Industry: A Story of Success in the Making

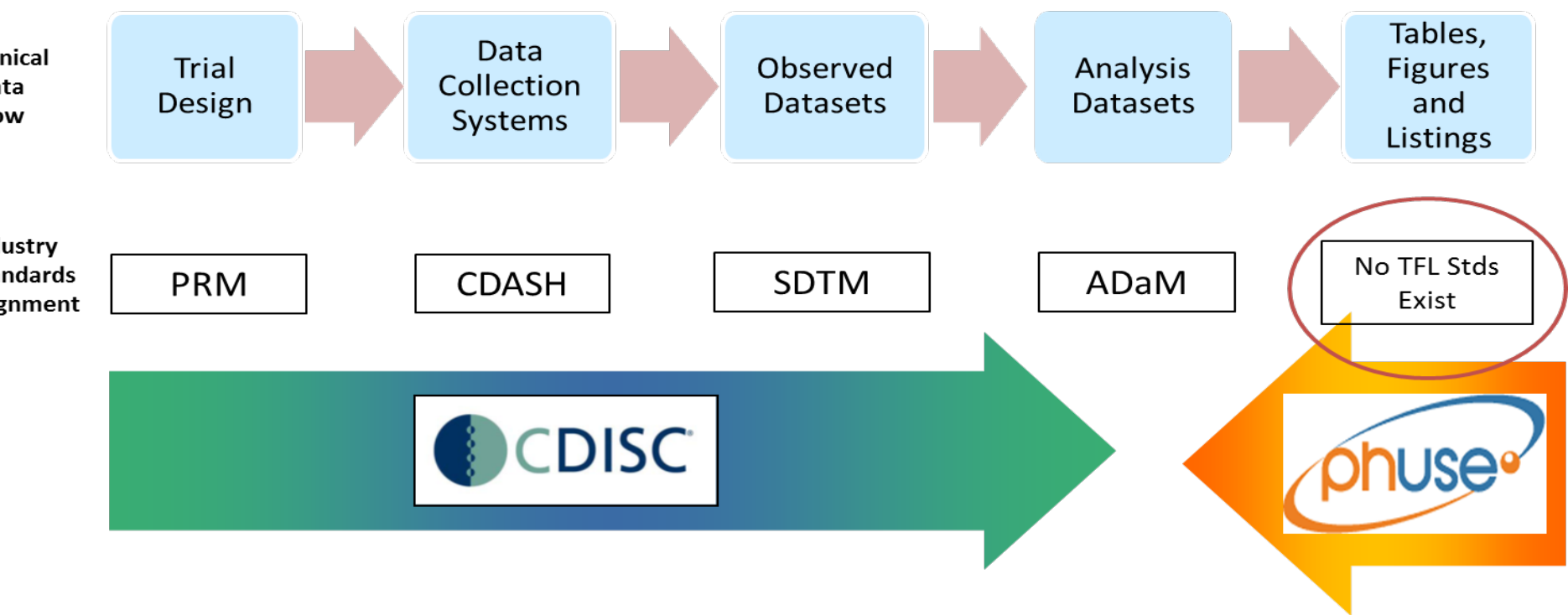
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ABSTRACT

Industry standards have evolved over time for data collection (CDASH), observed data (SDTM), and analysis datasets (ADaM). Although substantial progress has been made, additional standardization can improve product development. Development of standard tables and figures with associated analyses will lead to improved product life-cycle evaluation by ensuring reviewers receive the desired analyses for the evaluation of patient safety. More importantly, having an organized process for shared learning of improved methodologies can lead to earlier safety signal detection and better characterization of the safety profile of our products. A cross-industry working group (the PhUSE Computational Science Symposium Development of Standard Scripts for Analysis and Reporting Working Group) is providing recommendations for analyses, tables, and figures for data that is common across therapeutic areas (laboratory measurements, vital signs, electrocardiograms, adverse events, demographics, medications, disposition, hepatotoxicity, pharmacokinetics). A Script Repository has been created to contain associated code for the recommended analyses and displays. This poster will provide an update of this effort, and instructions for how to participate in the development and review process. We encourage all medical researchers to adopt this collaborative culture change!

SUMMARY OF STANDARDIZATION



PhUSE WHITE PAPER SUMMARY

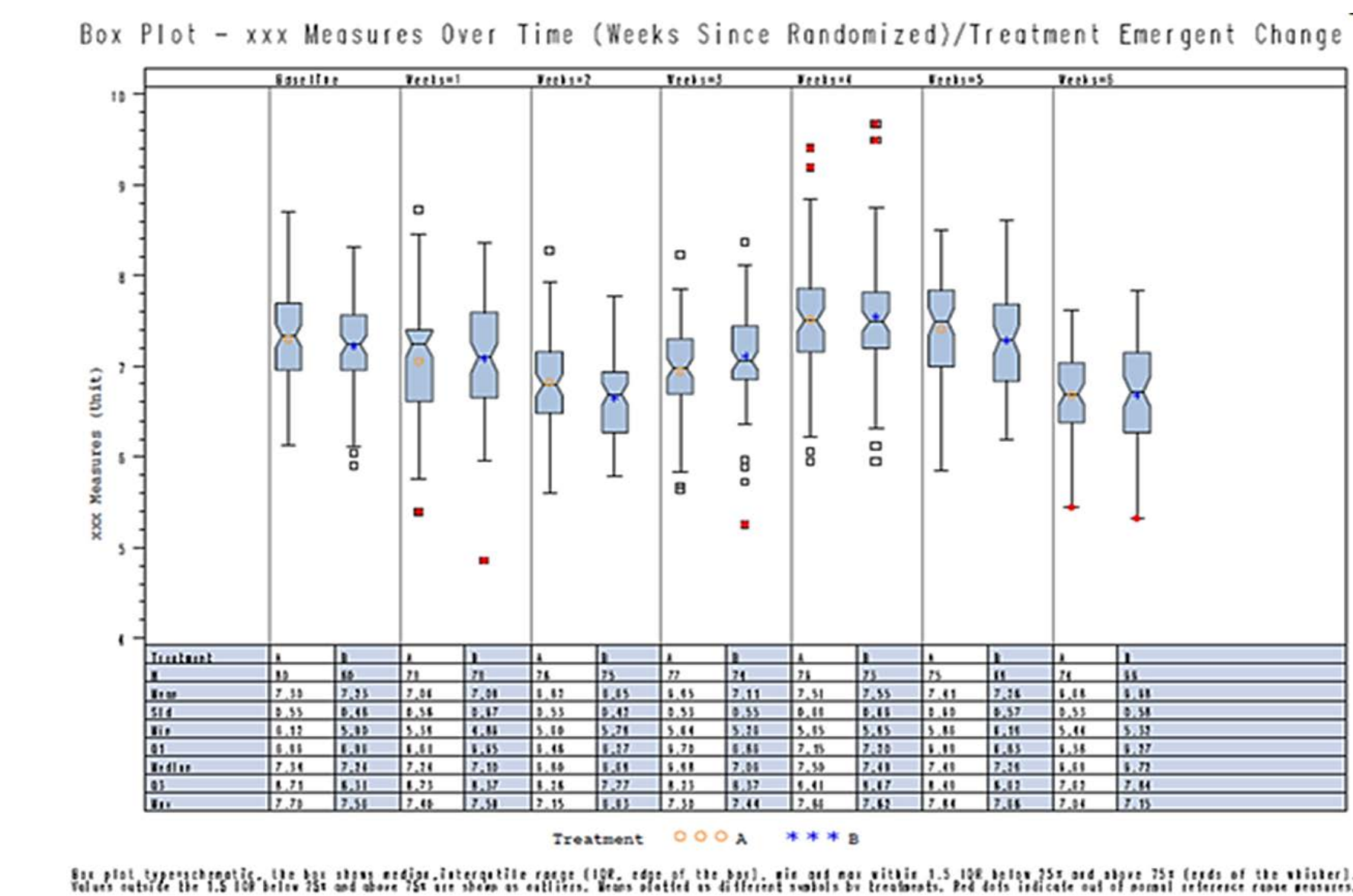
- Seven white papers at various stages
 - ECGs, Vitals, Labs – Central Tendency
 - Finalized October 2013
 - ECGs, Vitals, Labs – Outlier/Shifts
 - In progress – Target end of 2014
 - Adverse Events
 - In progress – Target end of 2014
 - Demographics, Disposition, Medications
 - In progress – Target Fall 2014
 - Hepatotoxicity
 - In progress – Target end of 2014
 - Pharmacokinetics
 - Finalized March 2014
 - QT Studies
 - In progress – Target early 2015

Final White Papers: www.phuse.eu
Draft White Papers: www.phusewiki.org

EXAMPLE DISPLAYS FROM THE WHITE PAPERS

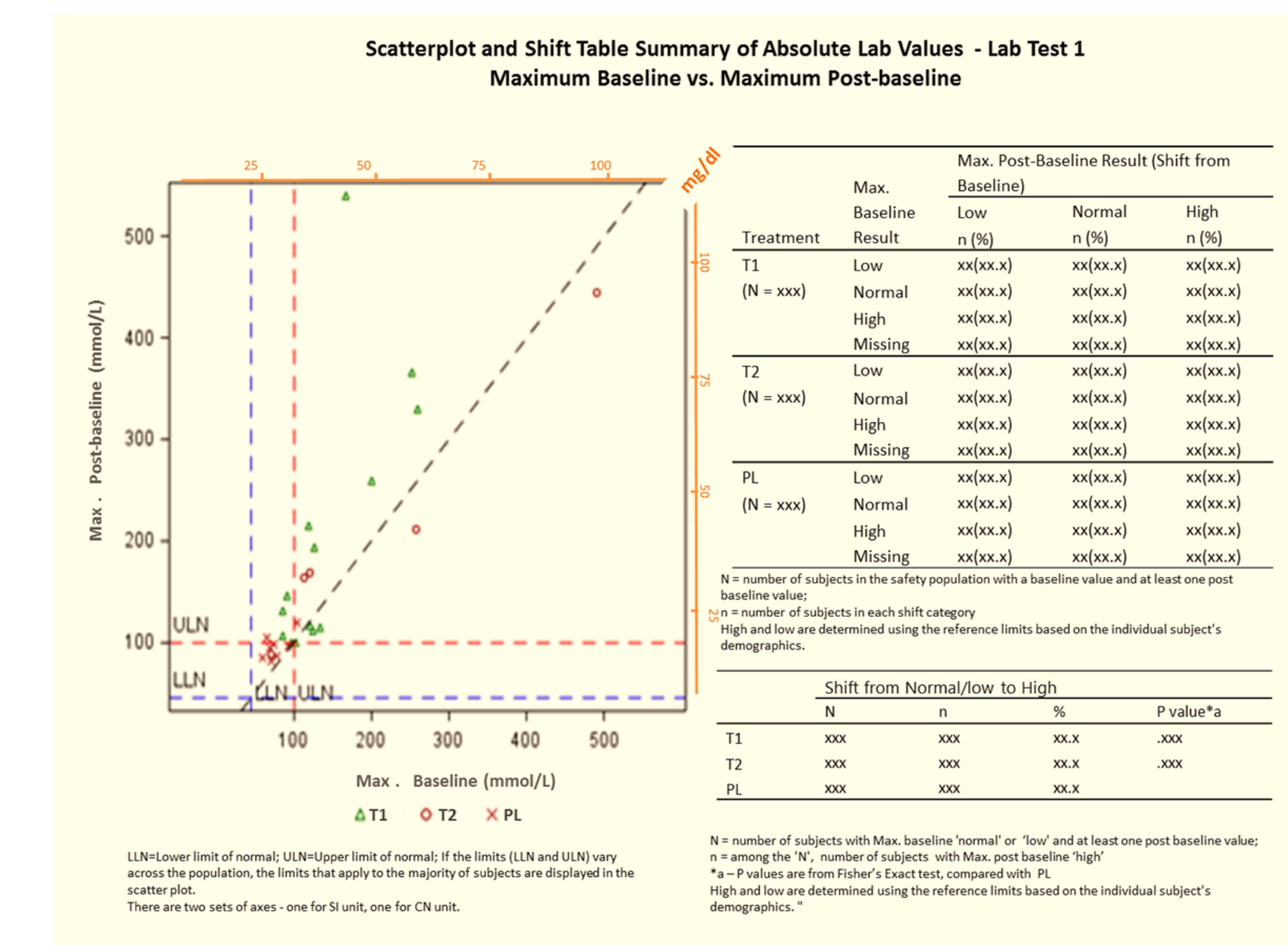
Mean Changes–Boxplots

- Allows for visual inspection changes over time
- Can visually assess the potential impact of outliers on the central tendency summary statistic
 - Out of range values in red
- Easy to see treatment differences
- Summary table complements box plot so numbers are available for textual summaries.

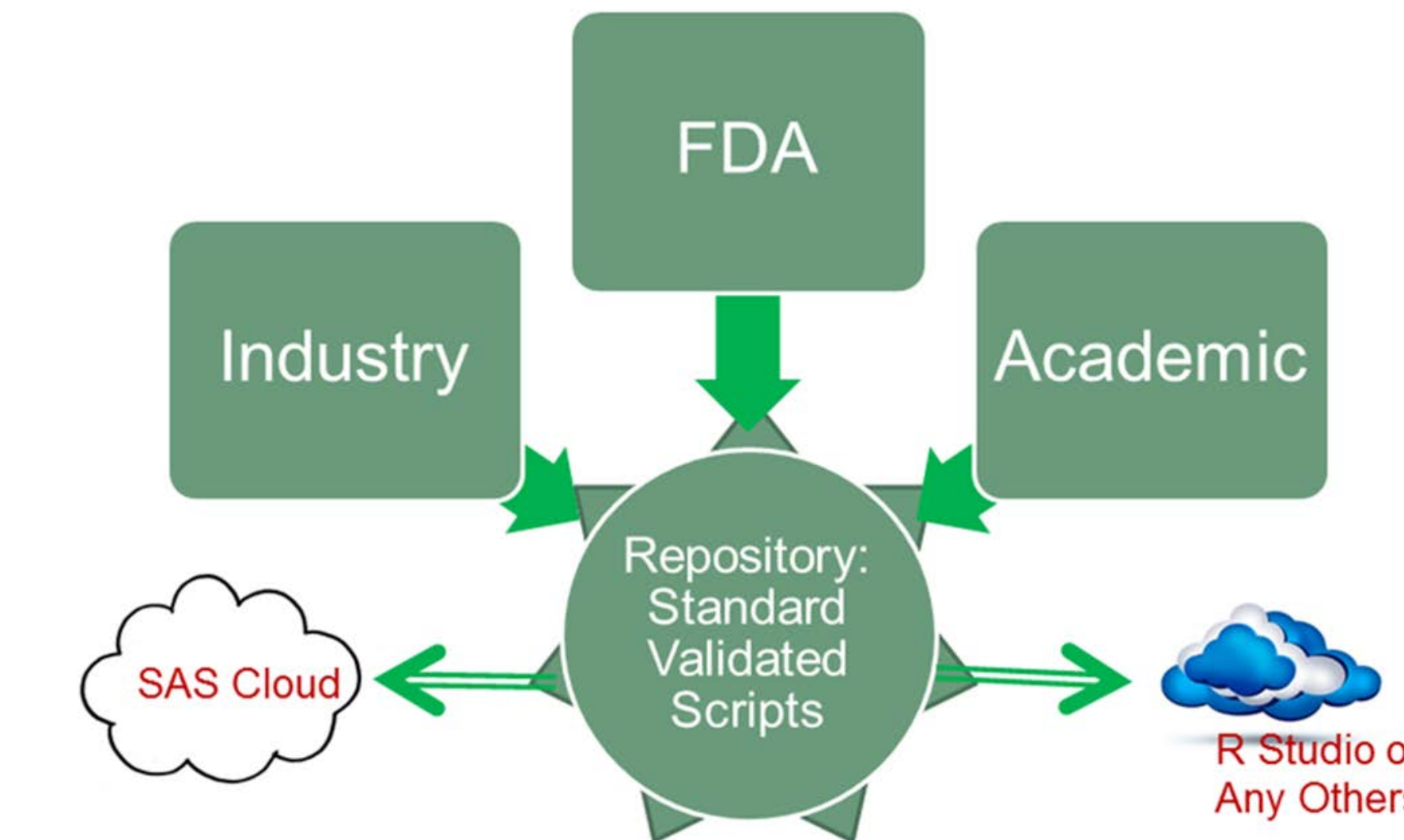


Outliers/Shifts

- Outliers/shifts analysis for numeric safety measures
 - Labs, ECGs, and Vital Signs
- Two pages per laboratory analyte
 - One for Max. baseline vs. Max. post-baseline
 - Another identical one for Min. baseline vs. Min.post-baseline
- Scatter plot
 - Shows patient-level information
 - Allows quick browsing through a large amount of information

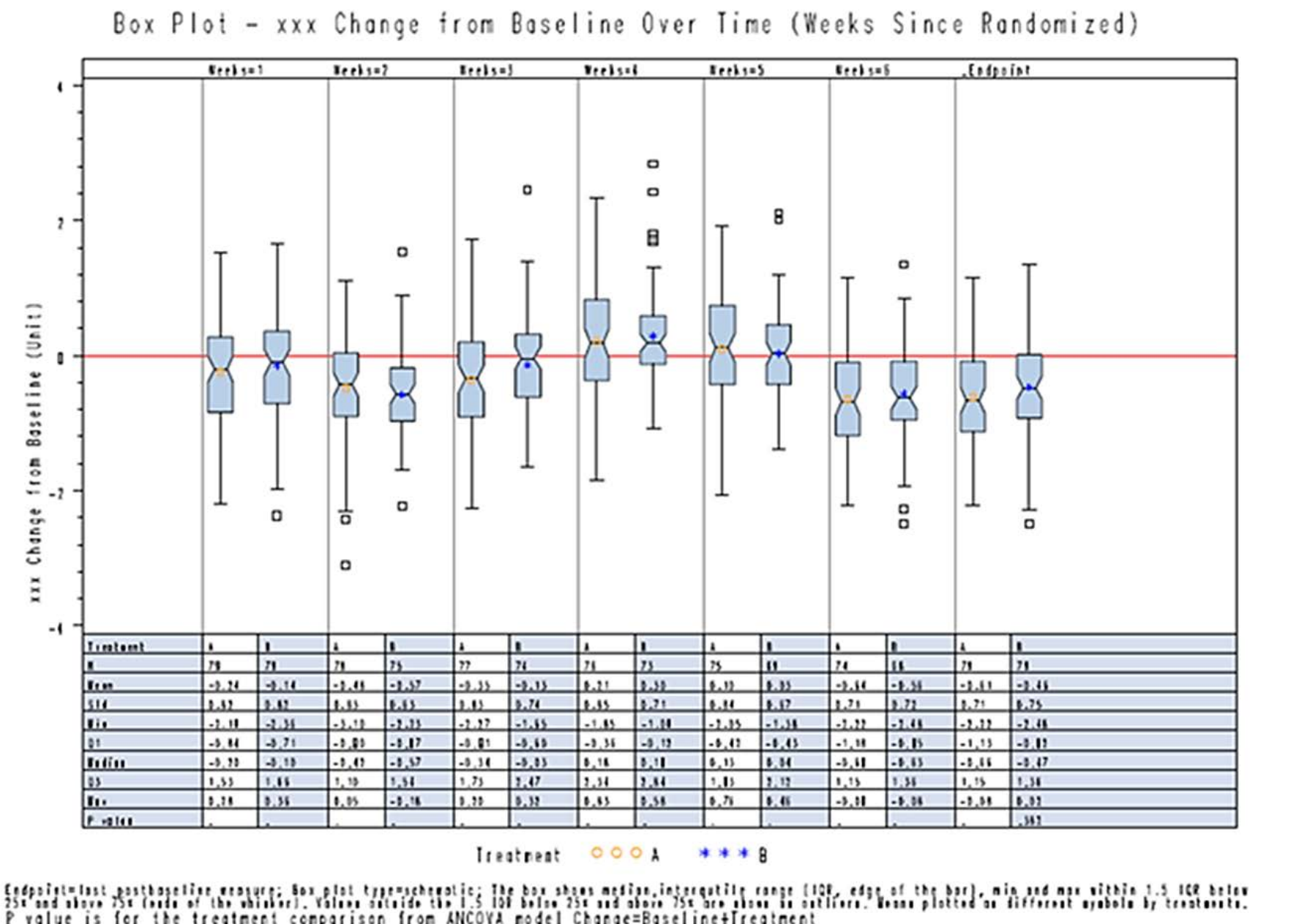


SCRIPT REPOSITORY – THE VISION

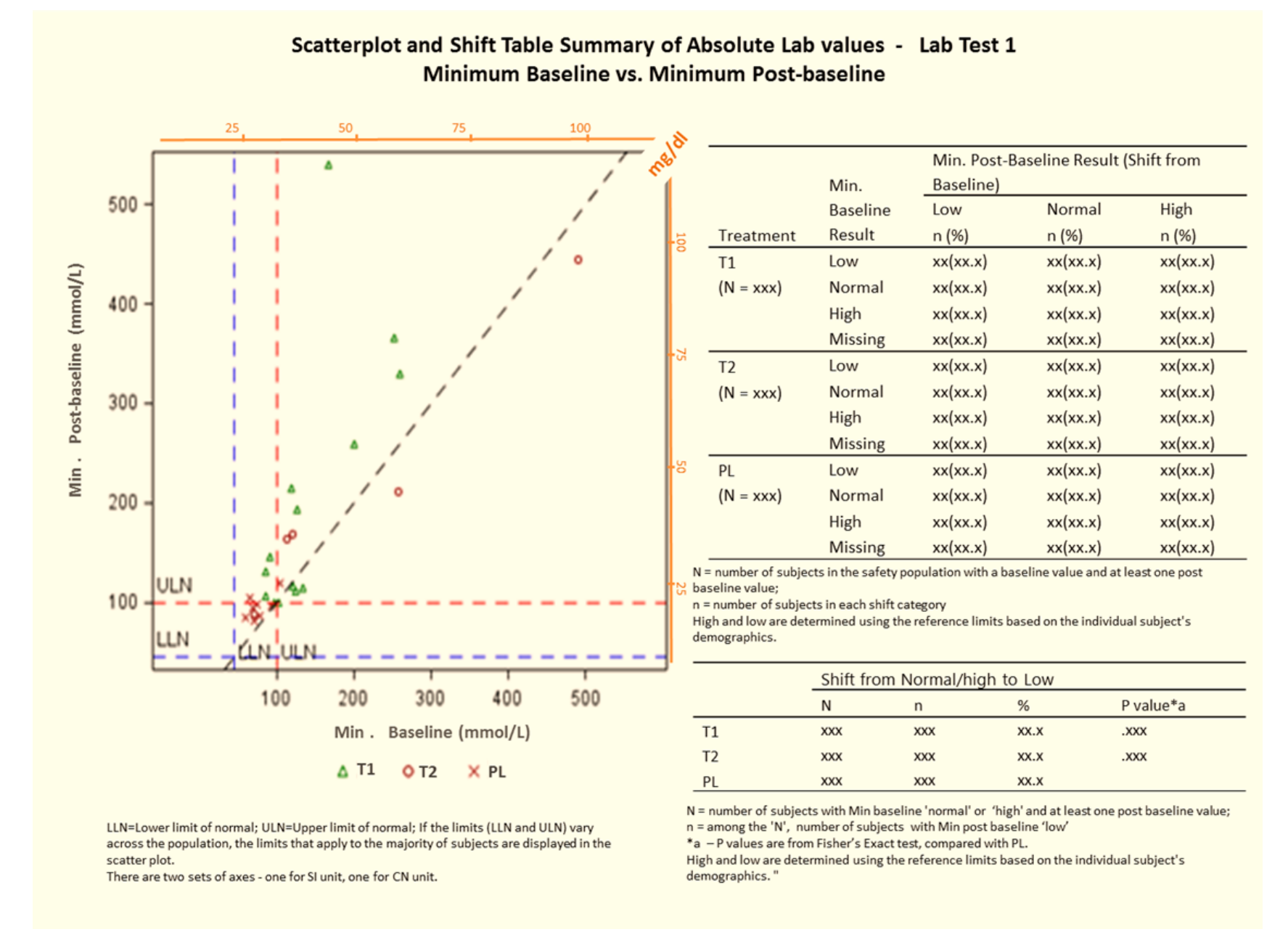


SCRIPT REPOSITORY

- PhUSE Repository:
 - <http://code.phuse.com>
 - <https://code.google.com/p/phuse-scripts/>
- Selected Google Code to host the standard script repository
 - Some scripts exist in the repository
 - MIT license chosen
 - Process guidelines developed
- Established the basic structure and process
 - Folder structure and name conventions
 - Roles and responsibilities for each role
 - Tasks and duties
 - Process of tracking issues
 - Required metadata and recommended programming style for scripts
 - Test data and validation documentation



- Shift table
 - Shows useful summary-level statistics
- Treatment-emergent high/low table
 - Shows concise summary-level statistics
 - Allows for easy assessment of treatment difference



Disposition Table

Subject Disposition	Treatment A (N=XXX) n (%)	Treatment B (N=XXX) n (%)	Total (N=XXX) n (%)	P-values ^a
Completed	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Death or Adverse Event	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Death	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Adverse Event	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Lack of Efficacy-Related Reasons	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Lack of Efficacy	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Progressive Disease	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Disease Relapse	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Other Reasons	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Lost to follow-up	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Non-compliance with Study Drug	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Pregnancy	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Protocol Violation	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Physician Decision	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Withdrawal by Subject	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Withdrawal by Parent/Guardian	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Recovery	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Technical Problems	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx
Other	xx(xx.x)	xx(xx.x)	xx(xx.x)	xx.xxx

Abbreviations: N = total number of subjects; n = number of subjects in the specified category
% = Percentage of patients
^aa = Fisher's exact test when comparing 2 treatment groups

CONCLUSIONS

- Industry standards have evolved over time for data collection (CDASH), observed data (SDTM), and analysis datasets (ADaM).
- Development of standard tables and figures with associated analyses is the next step.
- Having an organized process for shared learning of improved methodologies can lead to earlier safety signal detection and better characterization of the safety profile of our products.
- We welcome new members! Contact information on www.phusewiki.org